

Syngas/hydrogen production from self-supporting carbon sacrificial anode-assisted water electrolysis: Aspen Plus simulation

What is the gas composition of H₂ and CO stream?

Property Method: - ELECNRTL

Components: - Conventional: C, KOH, H₂, O₂, H₂O

Block Details:

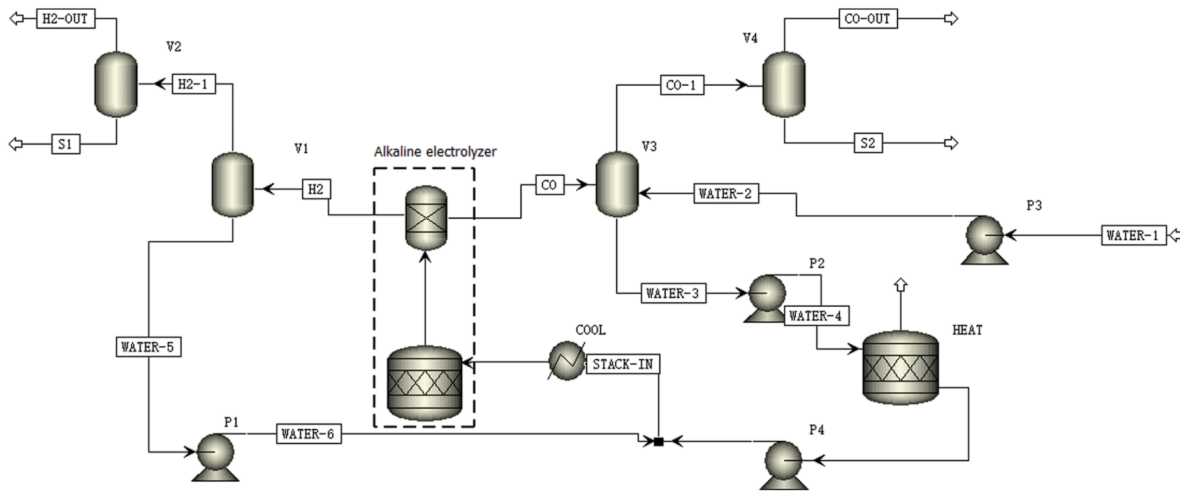


Fig. 1. Aspen simulation system of carbon-assisted water electrolysis designed for syngas/H₂ production.

Equations:

$$n_{H_2,prod} = \eta_F \times \frac{I}{z \cdot F} \times N \quad \dot{n}(H_2) = \frac{W_{Stack}}{V_{Cell} \times z \times F}$$

$$W_{stack} = V_{stack} \times I = (V_{cell} \times N) \times (i \times A_{cell})$$

$$V_{cell} = V_{rev} + [(r_1 + d_1) + r_2 T + d_2 p]i + s \times \log \left[\left(t_1 + \frac{t_2}{T} + \frac{t_3}{T^2} \right) i + 1 \right]$$

$$U = U_{rev} + \frac{r}{A} I + s \log \left(\frac{t}{A} I + 1 \right)$$

Item	Coefficient	Symbol	Value
1	V_{rev}	V	1.23
2	Current density	A/m^2	100, 500, 1000, 2000
3	Process temperature	$^{\circ}C$	75
4	r_1	$\Omega \cdot m^2$	4.45153×10^{-5}
5	r_2	$\Omega m^2/^{\circ}C$	6.88874×10^{-7}
6	F	C/mol	96,485
7	s	V	0.33824
8	t_1	m^2/A	-0.01539
9	t_2	$m^2/^{\circ}C/A$	2.00181
10	t_3	$m^2/^{\circ}C^2/A$	15.24178
11	V_{cell}	V	1.36, 1.56, 1.67, 1.81
12	A_{cell}	m^2	0.1

Proximate analysis wt. %				Ultimate analysis wt. %				
Mar	FC _{ar}	V _{ar}	A _{ar}	C _{ar}	H _{ar}	O _{ar}	N _{ar}	S _{ar}
10.40	14.50	73.30	1.80	41.31	5.55	40.37	0.42	0.15

(Assume all the required data.)